Data Analysis Project – Items to Think About

At this point, hopefully you've engaged in some exploratory data analysis for the project. Here are some things to bear in mind as you head into the modelling phase.

1. There are many ways to tackle this project in terms of what models you can fit. At a minimum, you need an appropriate tree and an appropriate GLM that show some evidence of being arrived at via a process. I.E. not the default tree and kitchen-sink GLM without further process. Convince the client they made a good decision hiring you and that you did the work to get them the best model you could!

2. What are appropriate GLMs and trees? This depends on your response variable. **What response variable are you using?** Read carefully. We aren't predicting "price". We are predicting whether or not price is greater than half a million dollars. There are ways to predict "price" and then convert everything into the "over" or "under" idea, but to do the tree and GLM, use the binary response!

3. Check that every variable is being treated by R in a manner that makes sense. (e.g. are numerically coded categorical variables being treated as numbers or factors?) This should be done before you fit anything. If they aren't, when R treats them as numbers, it assumes 1 unit increases are meaningful. For example, if a variable has levels 1 through 5, and you treat it as a number, then 4 is really twice 2, rather than being just 2 levels different.

4. Consider variable relationships among predictors. You want to avoid rank deficient matrices / linear dependencies. Discuss your variable selection.

5. Consider useful re-expressions of predictors (even if they don't end up in your final model). I.E. Don't just assume that you should or have to use the predictors as they are presented to you. This may be useful in situations where you see predictors that don't have a lot of variability – perhaps a presence/absence or other indicator could be more useful than the original variable itself.

6. Latitude and longitude are present in the data set. Even if you don't overlay the observations on an actual map, you can make plots to examine variables with lat/long as the grid. Should these be predictors themselves?

7. Be sure you use a training/test and CV framework that allows you to compare models. For example, don't make a training/test set for the GLM and a different training/test set for the tree. Similarly, you can't compare CV results unless the models have the same framework.

8. Use the power of the tidyverse and ggplot2 for visuals. The results look much nicer than base R commands.

9. Be sure your model output looks correct / doesn't have anything strange or a warning with it that you don't expect.

10. Be sure to give a reasonable sense of how your final model is doing. What is a good baseline to compare to?